

Astro326 Lab 3

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1. Abstract

In this paper we will look at the Cassiopeia A and the pulsar B0329+54 through the lens of the Algonquin Radio Observatory. We will investigate the properties including the optimal frequency at which measurements are done for the telescope with Cassiopeia by looking for the gain and noise. We will then look at the pulsar to find the pulse profile.

2. Introduction

Radio astronomy enables the study of celestial objects by detecting and analyzing radio waves they emit. This lab will focus on analyzing radio observations from the Algonquin Radio Observatory (ARO) which is Canada's largest single dish telescope. We will observe two data sets that were taken there. The first focusses on the supernova remnant Cassiopeia A (Cas A) and the second are observations of the pulsar B0329+54. With these we will be able to both test the telescope's performance using beam width and look at some of the properties of these 2 objects.

Cassiopeia A is one of the brightest radio supernova remnants in the sky, making it an ideal target for analysis of emission profile. The Pulsar B0329+54 is also one of the brightest pulsars so is an ideal candidate when studying periodic signals. Both of these objects will be looked at further later in the report. We will use both of these to identify characteristics of the telescope such as beam width, effective aperture, and system temperature. These will be estimated using various methods, such as Gaussian fits or curve fits and will help us understand both advantages and limitations of the ARO.

The lab will help us understand and test some of the key techniques commonly used in radio astronomy through some of the most widely known and used astronomical objects. By analyzing both a supernova and pulsar, we can explore 2 fundamental aspects of radio astronomy, the performance and the time analysis respectively.

3. Data and Observations

As stated above, we will look at 2 different data sets, both taken from ARO telescope.

3.1. Algonquin Radio Observatory (ARO)

The ARO is a radio observatory has the largest radio telescope in all of Canada, spanning about 46 meters. It is located in Algonquin Provincial Park, Ontario, Canada. The ARO 46 meter telescope is capable of high-sensitivity observations across various radio frequencies. When used in conjunction with the 26 meter telescope in British Columbia, it simulates a 3074-kilometre radio telescope. (Astrolab). For this lab, we focus exclusively on the ARO 46-meter telescope, utilizing its observations of Cassiopeia A and Pulsar B0329+54 to study telescope beam characteristics, noise properties, and pulsar signals.

3.2. Cassiopeia A

The Cassiopeia A is one of the most studied super nova remnants in space. It is located about 11 000 light years away from earth in the Cassiopeia constellation. The explosion roughly happened 300 years ago but has not yet reached earth due to its distance. (Cassiopeia A) In this report we will use it to identify the beam width and effective aperture of the ARO telescope by studying the radio emissions of the nebula. The figure 1 below shows the raw data taken from the ARO data set and converted into a waterfall plot.

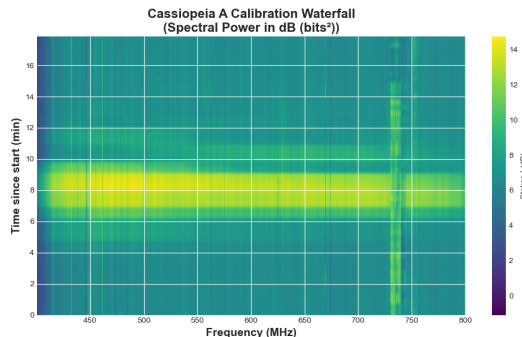


Fig. 1.—: Cassiopeia waterfall Plot

3.3. Pulsar B0329+54

Pulsar B0329+54 is located approximately 3460 light years away from earth and about 1.97 ± 0.19 the mass of earth. (Wikimedia). It is one of the most studied pulsars making it a good target for us to study and use for nits pulse profile. We will use folding techniques and to enhance the

signal to noise ratio and investigate the pulsar’s periodicity stability. In figure 2 below, we have read out and put the raw data of the pulsar into a water fall plot. As we can see there is not much to analyze at the moment.

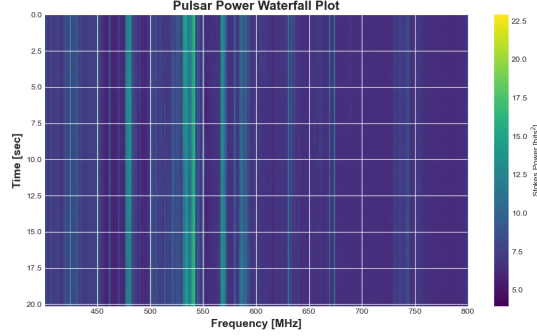


Fig. 2.—: Pulsar B0329+54 waterfall plot

3.4. Normal/Guassian distribution

The equation for the normal distribution, otherwise know as the gaussian distribution is a statistical method that is symmetrical around the mean of data. The equation below represents is used for the distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad (1)$$

The distribution forms a bell curve around the mean. This means there is a high concentration of points around the mean and as you move farther from it (in both directions) the density gets smaller to. We can see this through the standard deviations: approximately 68% of values fall within one standard deviation of the mean, 95% within two, and 99.7% within three, making this a very useful model for natural variations. (Cowan 2023) Additionally, the accuracy of the Gaussian distribution is more accurate the more points there are.

3.5. Fitting

e will be using the Linear least-squares fitting which describes the statistical method that looks a set of data point and find the best fit. It does this by minimizing the distances between the points and the linear fitted line. The linear fit is shown below:

$$y = mx + b \quad (2)$$

The Least Square fit is below: For a polynomial:

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

To find the residuals of this:

$$\chi^2 = \sum_{i=0}^N \frac{(y_i - f(x_i))^2}{\sigma_i^2}$$

Where the y_i are the observed value, $f(x_i)$ the polynomial predictions, and σ_i the uncertainties. In Python, we will utilize the `np.curvefit` function to do this for us.

4. Data and Reduction

In this section we will discuss how we reduced and manipulated the data set to have better diagrams. We did not make many reduction in the Cassiopeia data set apart for a 15% error margin for the effective aperture.

For the Pulsar, we will use de-dispersion and signal folding to manipulate and amplify the signal of the pulsar that we want.

5. Data analysis

In this section, we will manipulate and analyze the data of the both Cassiopeia and the Pulsar through various methods.

5.1. Cassiopeia A

By starting with the raw data of Cassiopeia, we first measure the received signal at one point, and found the power variation. We then redid this for every point to find the strongest signal among all the recorded ones. (Figure 3)

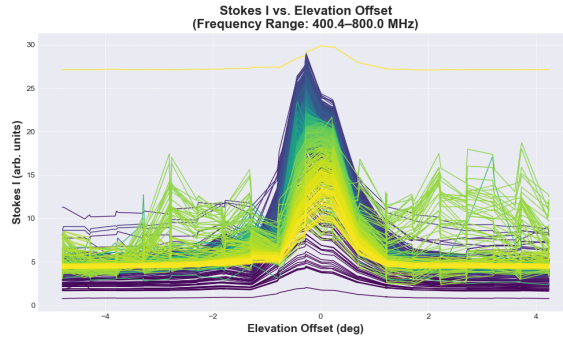


Fig. 3.—: Power variation with source offset for all frequencies during Cassiopeia A calibration.

Figure 4 and 5 look at the beam patter and the residual at 600.39 MHz of the Gaussian fitted on it. The Gaussian function can then provide both the full width at half maximum (FWHM), which

tells us the angular resolution. Finally, using this we can estimate the beam width at this frequency.

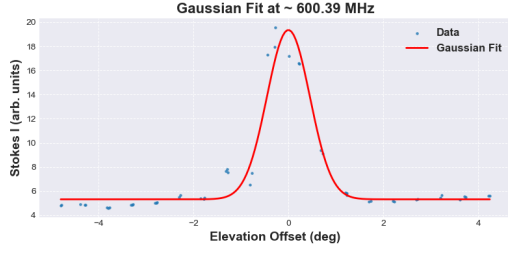


Fig. 4.—: Gaussian fit to the beam pattern of Cassiopeia A.

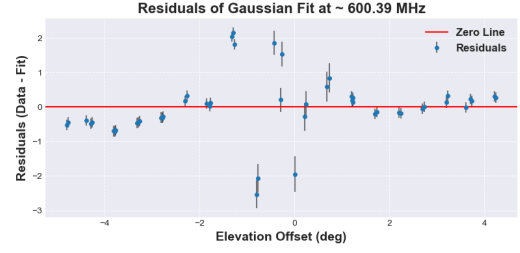


Fig. 5.—: Uncertainty model for the beam pattern of Cassiopeia A.

In Figure 6, by using the angular resolution, we can plot the Beam width and aperture efficiency against the frequency. We observe that there are some points which are outliers compared to the other points on the beam width. The effective aperture decreases as frequency increases. Which can be seen even more clearly in Figure 7. The outlier points are less visible but are still present for the effective aperture.

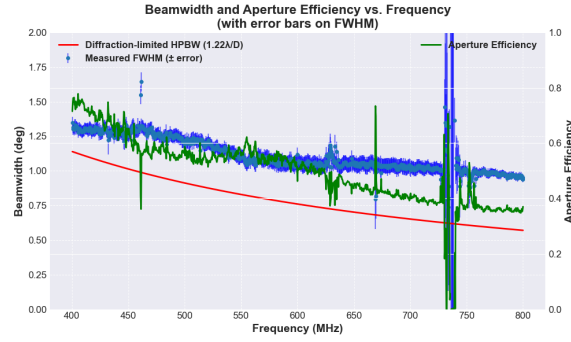


Fig. 6.—: Variation of beamwidth with frequency for Cassiopeia A.

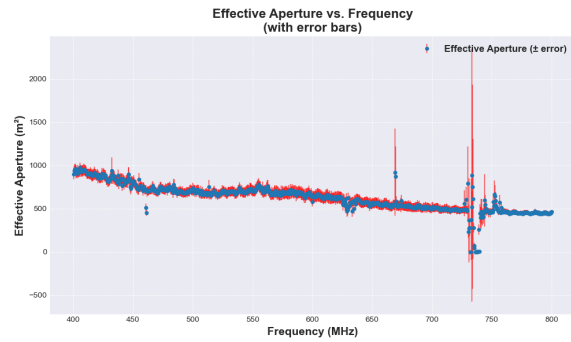


Fig. 7.—: Variation of effective aperture with frequency for Cassiopeia A.

Figure 8 and 9 show the receiver temperature vs frequency and the receiver gain vs frequency respectively. The temperature shows the noise contribution affecting the system, which we see is quite small as frequency increases. On the other hand, the Gain increases as the frequency increases and goes back down at around 750 MHz. So this shows that the telescopes works best at frequencies of around 700Mhz.

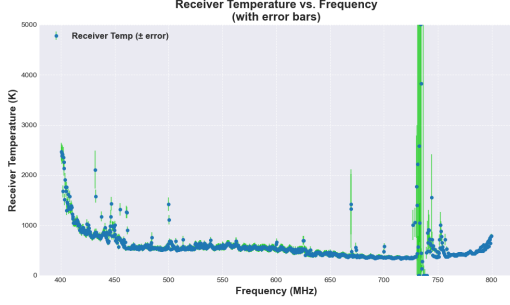


Fig. 8.—: Receiver temperature as a function of frequency, showing system noise contributions.

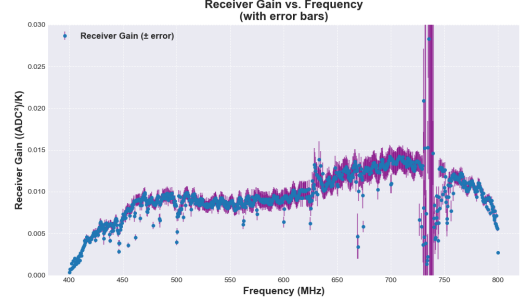


Fig. 9.—: Receiver gain as a function of frequency, highlighting variations in sensitivity..

5.2. Pulsar

We will now start analyzing our pulsar data starting from the figure 2, the waterfall plot of the raw data. We first Normalize the waterfall plot and did an initial RFI cleaning for better data. We then calculated the dispersive time delay caused by the interstellar medium (Figure 11) and plotted it to get the frequency-dependent delay for Pulsar. We then fold the data of the and replotted the waterfall graph, now showing a clear pulse profile for the pulsar (Figure 12).

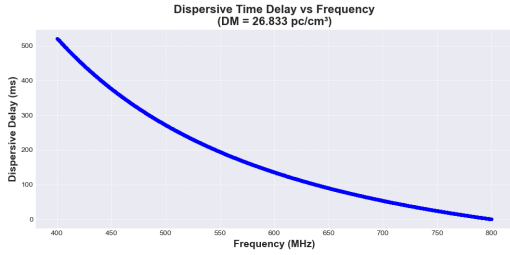


Fig. 10.—: Dispersive time delay vs. frequency for Pulsar B0329+54.

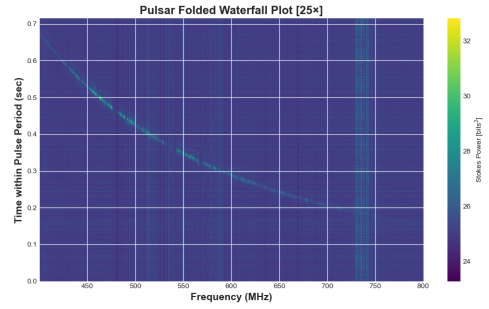


Fig. 11.—: Folded pulse profile of Pulsar B0329+54

By looking at figure 12, we see that the pulse is a curve when we want a straight line. To realign this, we will apply a de-dispersion algorithm, giving us figure 13.

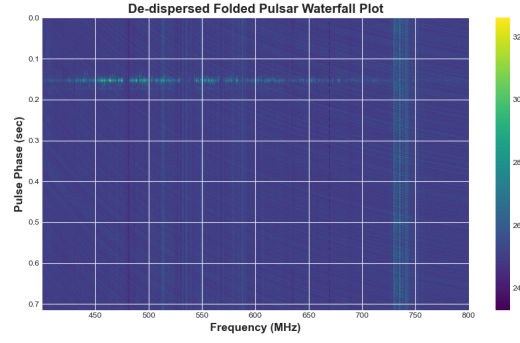


Fig. 12.—: De-dispersion of the Pulsar B0329+54 signal.

Looking over all the pulse signals (Figure 14), we were able to average them to get the average pulse profile across all the pulses. We can see that there is 1 large peak at a time of around 0.15 sec within the pulse signal and a much smaller peak at a time of around 0.18 sec.

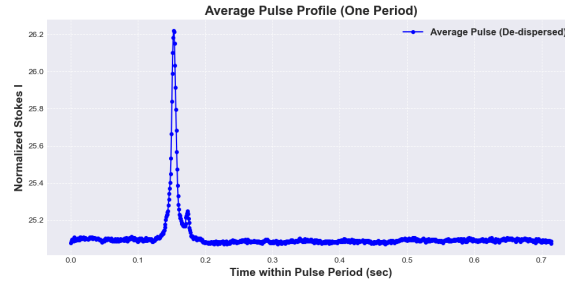


Fig. 13.—: Average pulse profile of Pulsar B0329+54 across all observed pulses.

The histogram in Figure 15 shows the amplitudes in a single pulse. This would imply that it is a strong pulse and we have a good baseline for our waterfall plot.

In plot 16 we look at the pulse timing stability. We can see that there are a lot of points that drop, which means that the calibration or limiting of data could have been done better. So, the consistency of the periodic emission is not very good.

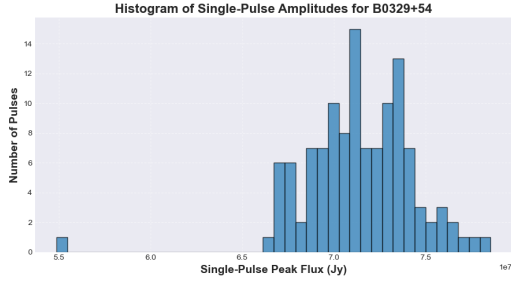


Fig. 14.—: Histogram of a single pulse from Pulsar B0329+54.

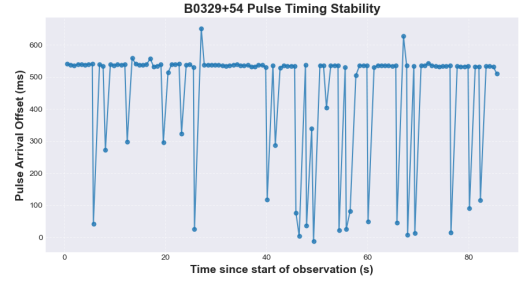


Fig. 15.—: Stability of arrival times for Pulsar B0329+54.

6. Discussion and conclusion

In this experiment, we utilized observational data from the ARO 46-meter telescope to study both Cassiopeia A and Pulsar B0329+54. The analysis of Cassiopeia A was able to help us understand what the optimal frequency for measurement with the telescope. We found that this was about 700 MHz as there were some outlier data points at around 750 and the rest of the points were slightly lower. We also were able to find that the aperture efficiency generally decreases as the frequency increases, further enforcing the reason of using a high frequency such as 700 MHz.

On the other hand, when analyzing the Pulsar B0329+54 we used various techniques to improve signal quality, such as de-dispersion and signal folding. However, the pulse arrival time stability was not consistent which was most likely due to our data limitations or our manipulations not giving us what we wanted. Despite this, we managed to use the folding and de-dispersion processes successfully enhancing the signal to find an average pulse profile for the pulsar.

While many sections of data manipulation were challenging, we were both able to find the noise caused by temperature, the receiver gain, and an optimal frequency for measurements. Future observations and refinements in data processing could improve the quality of the analysis, leading to more accurate conclusions about the behavior of both Cassiopeia A and Pulsar B0329+54. Overall, this experiment demonstrated the importance of understanding both telescope performance and the characteristics of astronomical objects in radio astronomy.

7. References

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8. Abstract